GDPS MGM 4-site Workshop - MGM Overview

GDPS

Providing world class protection

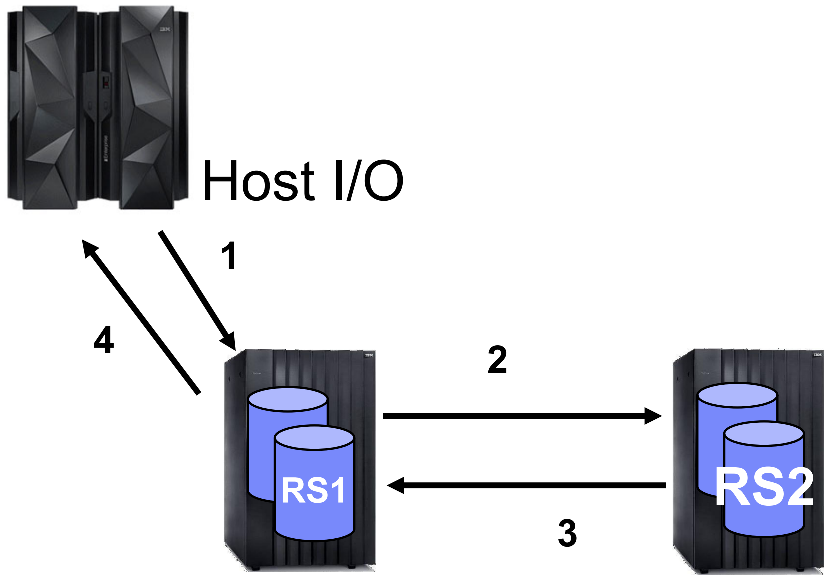
Session Objectives

- The purpose of this module is to provide information on the underlying remote copy technology that is used for MGM
- Information is presented in the following sequence:
  - PPRC Refresher
  - GM Refresher
  - Combining PPRC and GM to provide MGM
    - Includes:
      - What is cascade?
      - What is multi-target?
      - What is incremental resync?
      - Understanding the different remote copy relationships that are established to provide the MGM capability

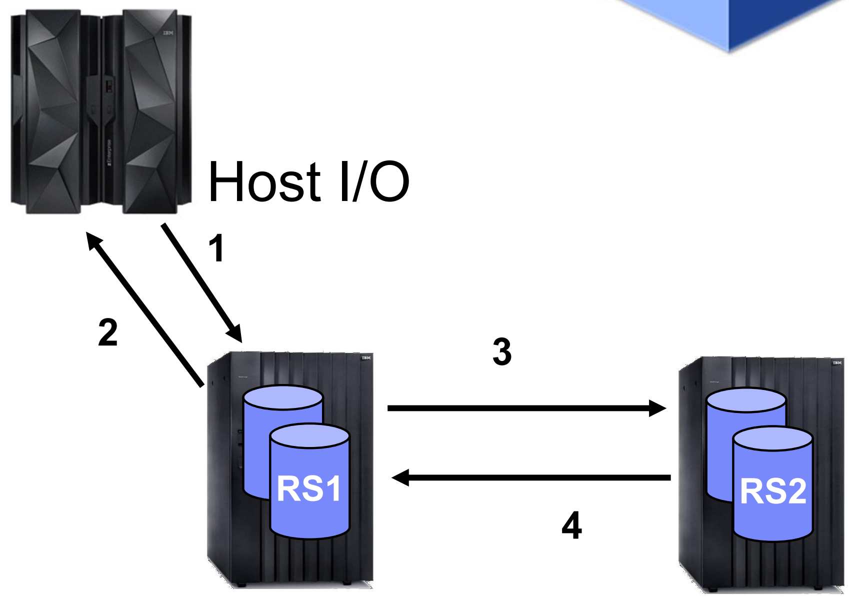
Pre-requisites & Restrictions

- Pre-reqs
  - You need to have the appropriate hardware, microcode levels and feature codes installed to be able to establish your MGM configuration
  - Consult with your storage technical specialist to understand the precise details
  - Disk must be multi-target capable
- Restrictions
  - Testing MGM-IR configurations with GDPS MGM requires at least two different disk subsystems (for the RegA.RS1 disk and RegA.RS2 disk)

Metro mirror and Global Copy



Metro Mirror



Global Copy

- Write to primary volume.
- The primary disk subsystem initiates an I/O to the secondary disk subsystem to transfer the data.
- Secondary indicates to the primary that the write is complete.
- Primary acknowledges to the application system that the write is complete.

- Write to primary volume.
- Primary acknowledges to the application system that the write is complete.

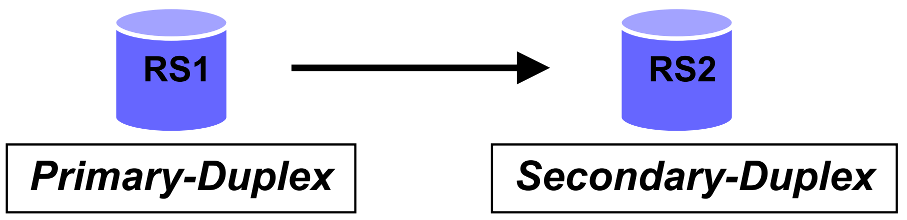
At some later time:

The primary disk subsystem initiates an I/O to the secondary disk subsystem to transfer the data.

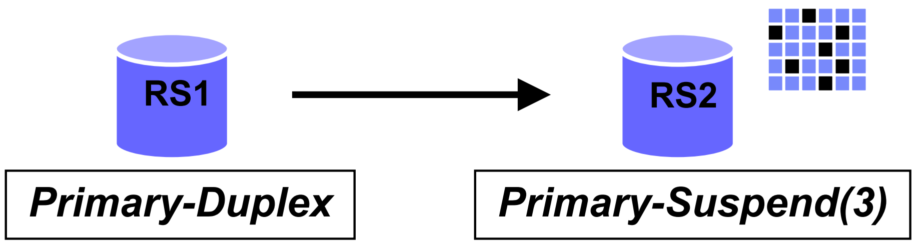
- Secondary indicates to the primary that the write is complete.
- Primary resets indication of modified track.

Failover/Failback refresher

- FAILOVER command used to make a secondary device a primary suspended device
  - Does not touch the 'original' primary device
  - Establishes a bitmap to track changes on the target device
- FAILBACK used to resynchronize the relationship
  - Can use either device in the suspended relationship as the new primary
  - Uses the bitmaps to determine what tracks must be copied to create the duplex pair



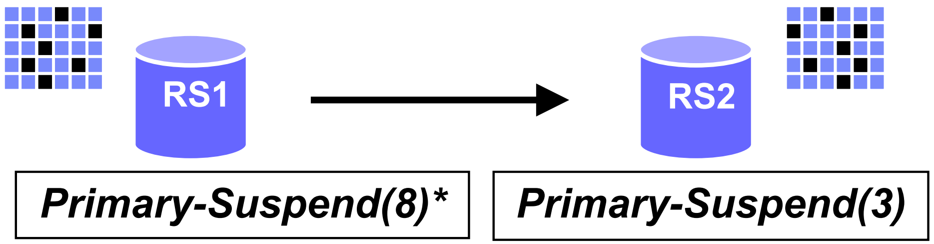
- **Normal operational mode**
- **Site A production site**
- **Site B recovery site**



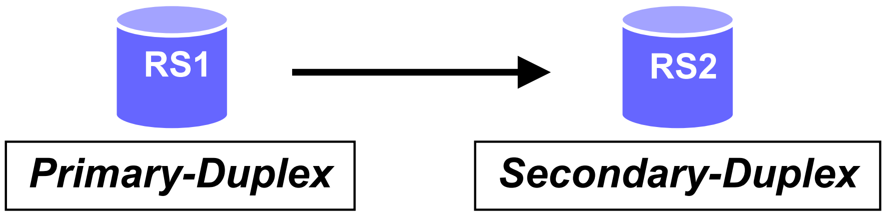
- **CESTPAIR DEVN(B) ACTION(FAILOVER)**
- **Change recording activated for B-disk**
- **B-disk is primary disk**
- **I/O can be performed against B-disk**
- **A-disk status will depend on situation**

FAILBACK A—B or B—A

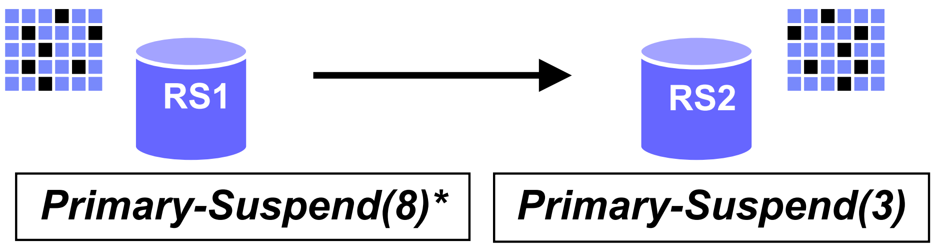
*** Pri-sus(8) below due to update to A after the failover of B**



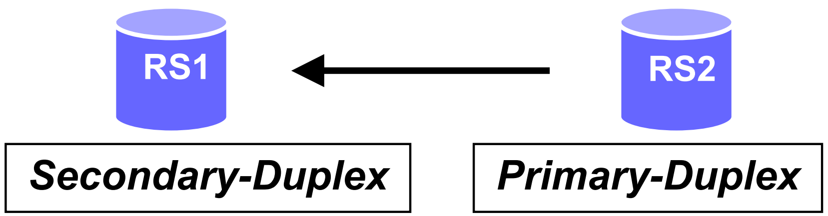
- CESTPAIR DEVN(A) ACTION(FAILBACK)
- OR A-disk and B-disk bitmaps
- Copy OOS tracks from A-disk to B-disk



- Full duplex relationship restored



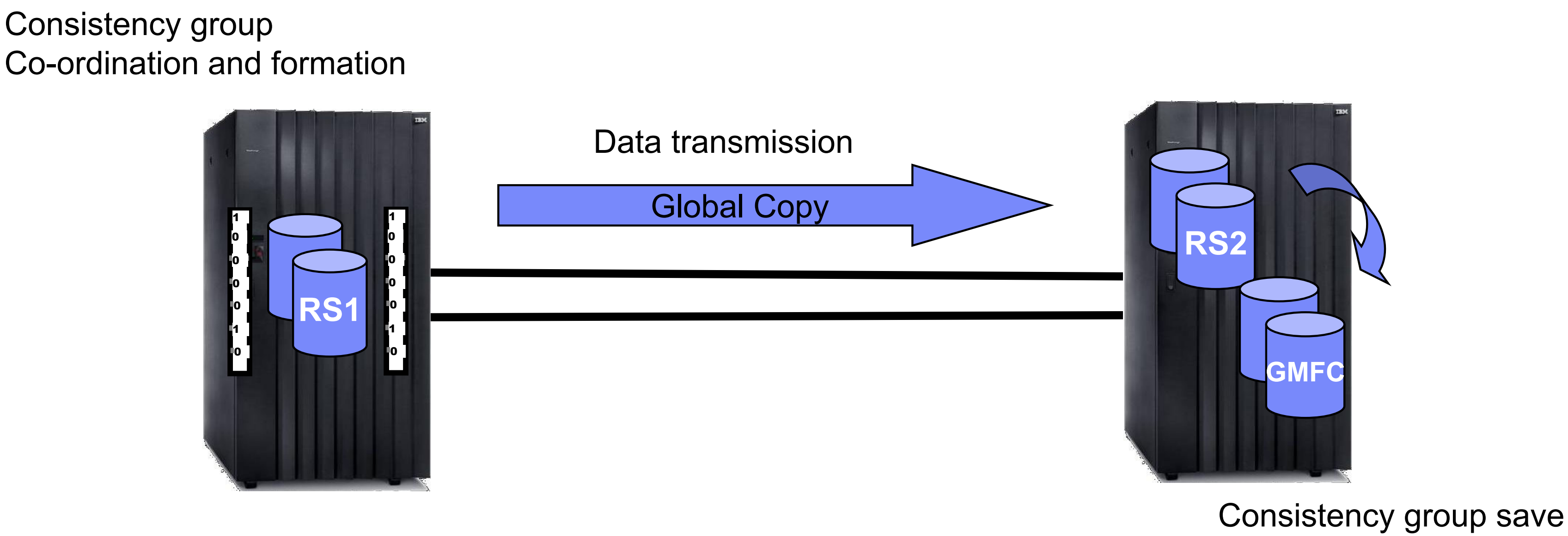
- Need to establish paths from B to A first
- CESTPAIR DEVN(B) ACTION(FAILBACK)
- OR B-disk and A-disk bitmaps
- Copy OOS tracks from B-disk to A-disk



- Full duplex relationship restored

Global Mirror

- Bitmaps used to hold consistency group information on primary disk subsystem
- PPRC paths used for disk subsystem to co-ordinate consistency groups among multiple disk subsystem
- Global Copy used to transmit consistency group between primary and secondary
- FlashCopy used to save consistency groups on secondary disk subsystem



A word on nomenclature

- We used to talk about Site1 disk, Site2 disk, A, B, C, D, GMFC, FC1 and so on but this became impossible to marry together for the multi-target and multi-solution environments over time for all the disk copies to be decribed, so we settled on an updated nomenclature based on regions and replication sites.

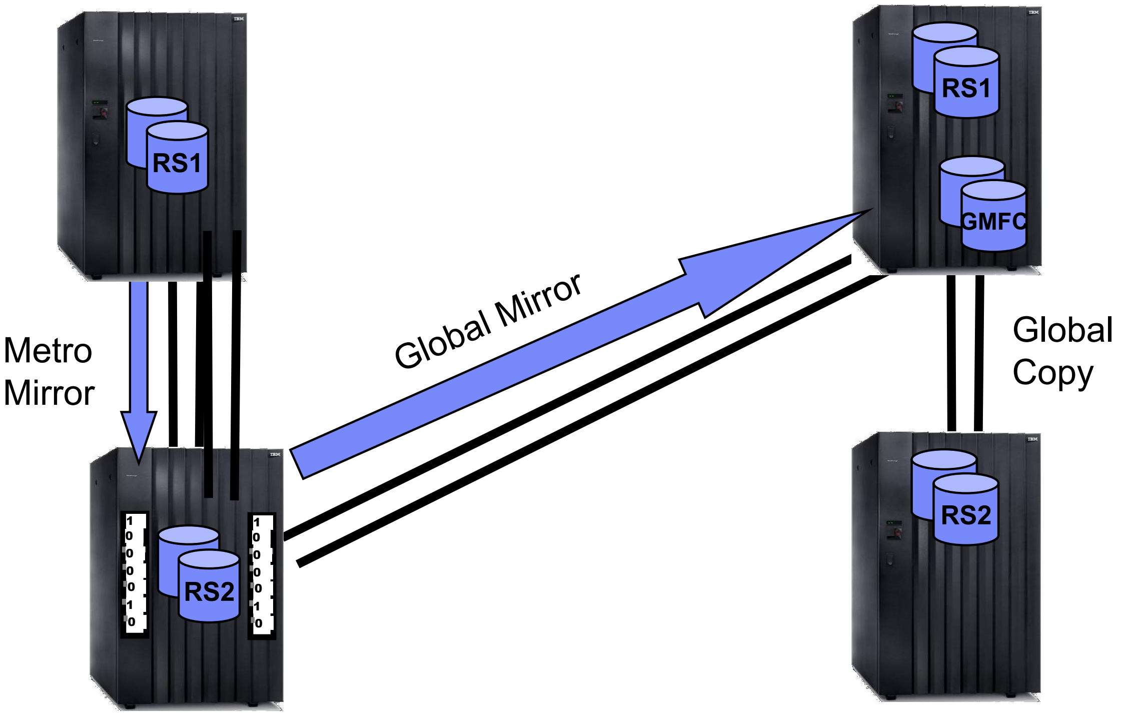
- Metro Mirror
  - Single Region solution, two or three copies of disk
  - Generally known as RS1, RS2 and RS3 (where multi-target)
  - Formally described as RegA.RS1, RegA.RS2 and RegA.RS3 but since a single region, the RegA is generally dropped.

- MGM 3-site
  - Two Region solution, three copies of disk
  - Formally described as RegA.RS1, RegA.RS2 and RegB.RS1. (For MGM 4-site, simply add RegB.RS2)
    - Describes the replication site (or copy) and the region
    - Region.GMFC used for the GM FlashCopy or Journals

- By default, the initial replication orientation for GDPS MGM is cascaded, not multi-target
  RegA.RS1 → RegA.RS2 → RegB.RS1 (previously ABC)

Metro Global Mirror

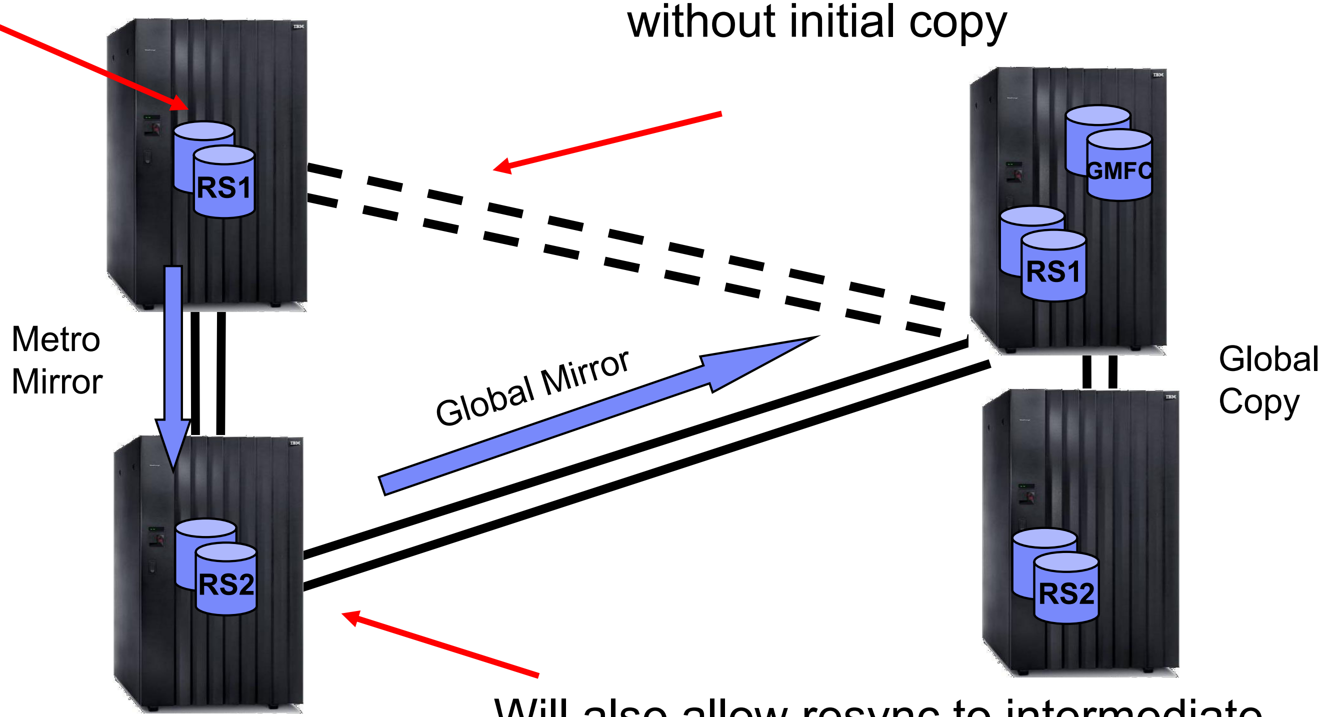
- Combines synchronous and asynchronous PPRC
  - Allows for local continuous availability with HyperSwap
  - Also provides out of region disaster recovery
  - Requires automation of PPRC environment to ensure consistent data at RegA.RS2
- Dual relationship on the RegA.RS2 disk – simplified with multi-target support
- RPO of 0 for RegA.RS1 site failure
  - Can recover to either RegA.RS2 or RegB.RS1
  - Zero RPO implies automation to ensure no production updates if mirroring stops
- Potential RPO of seconds for RegA.RS1 and RegA.RS2 twin site failure
  - Depends on workload and bandwidth between RegA.RS2 and RegB.RS1



What is MGM Incremental Resynchronization

Set INCRES on ESTPAIR to create bitmaps on PPRC primary disk subsystem to record differences between intermediate and remote

IR will allow resync from local to remote site in case of a intermediate disk failure without initial copy



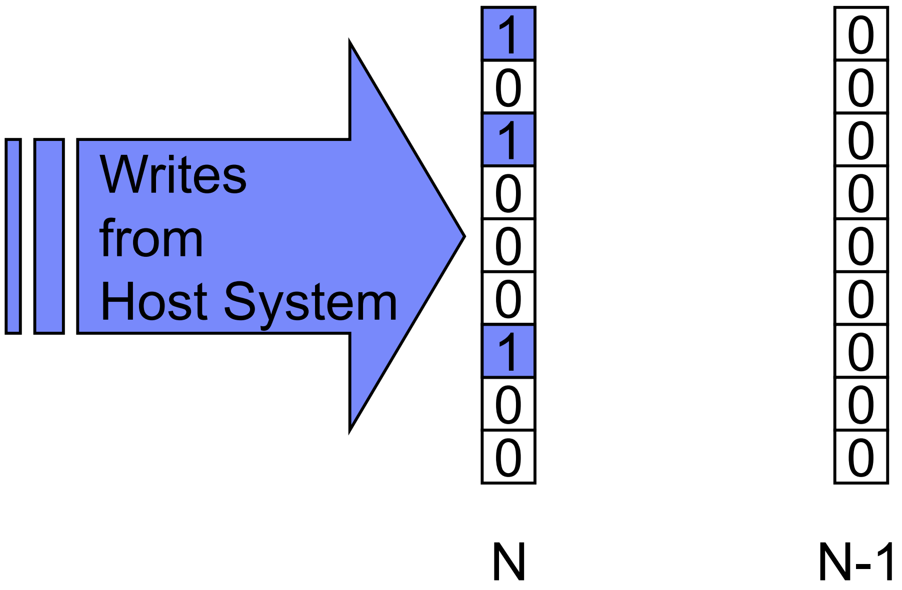
Will also allow resync to intermediate disk once the intermediate disk is restored

How does MGM IR work?

- Starts change recording on the PPRC Primary volume to keep track of the “changes in flight” to the Remote site in bitmaps.
- Coordinate Change Recording with Global Mirror consistency group formation. Toggle CR bitmaps (N, N-1) when a number of consistency groups have been formed at the Remote site.
- IR uses a conservative strategy that ensures consistency at the remote site if a resync (local to remote is needed).
  – Will result in more data being copied than absolutely necessary; but is safe!
- Primarily used when moving the GM session to the RegA.RS1 disk in the event of a failure of RegA.RS2 when running in a cascaded mode.
- When a resync is needed from RegA.RS1 to RegB.RS1, merge the CR bitmaps and the Out-of-Sync bitmap that is created when the PPRC relationship between RegA.RS1 and RegA.RS2 suspends. Send the “in flight” changes to the Remote site

How does MGM Incremental Resynchronization work?

PPRC primary box



Incoming writes from host recorded in “N” bitmap.

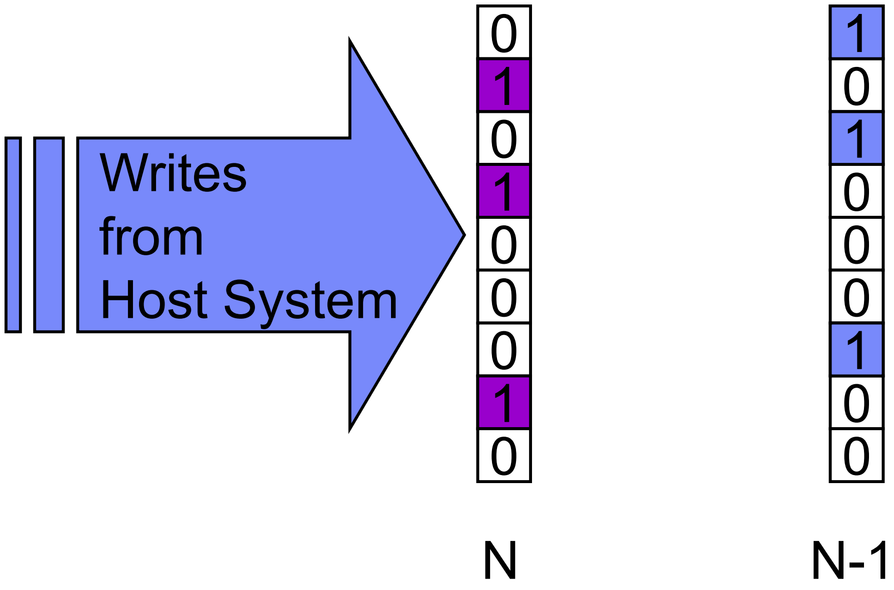
Periodically issue queries to the intermediate DS8K. If at least 3 CGs were successfully formed, and a minimum time has passed:

Toggle N, N-1 bitmaps

New writes still recorded in “N” bitmap

How does MGM Incremental Resynchronization work?

PPRC primary box



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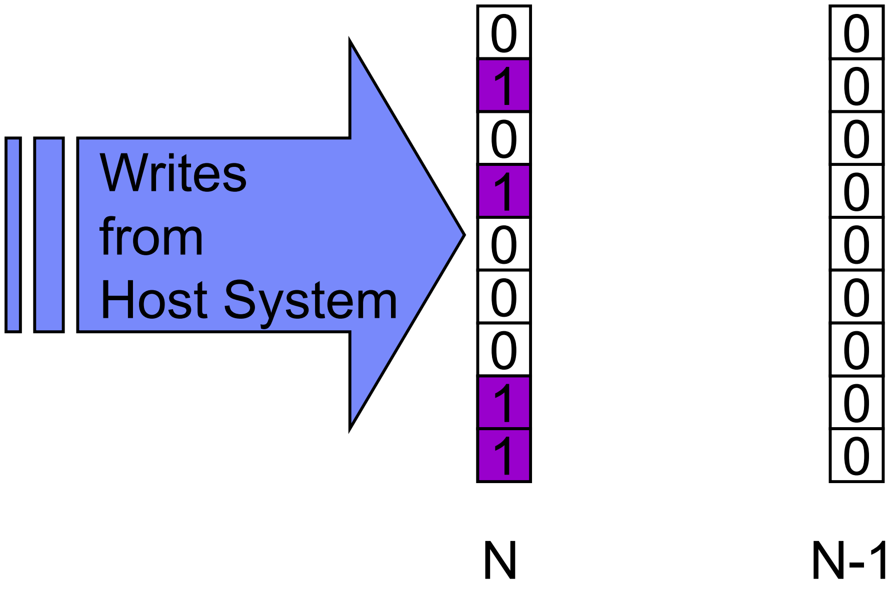


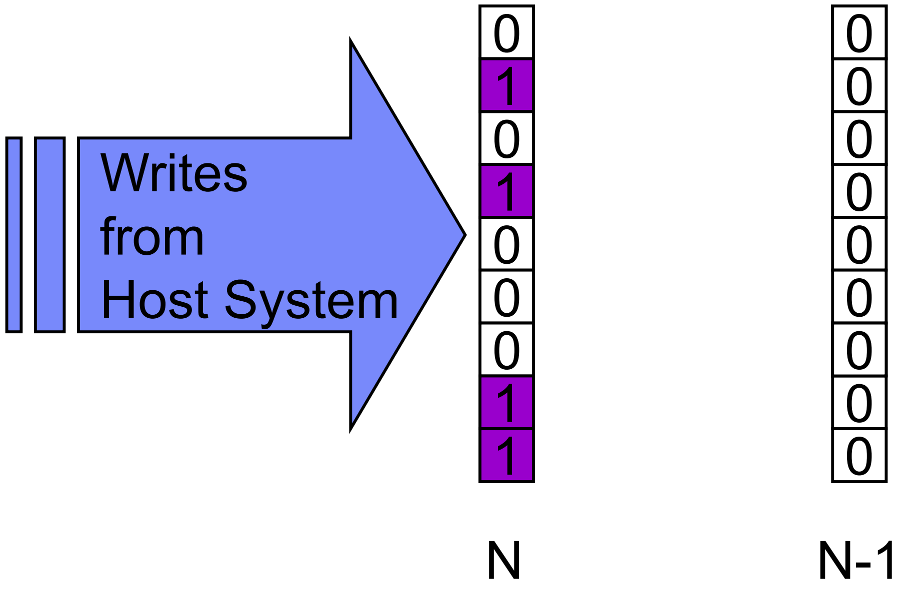
New writes still recorded in “N” bitmap

Issue queries to intermediate. If at least 3 CGs were successfully formed, and minimum time has passed:

Toggle N, N-1 bitmaps (N bitmap becomes zeros)

How does MGM Incremental Resynchronization work?

PPRC primary box



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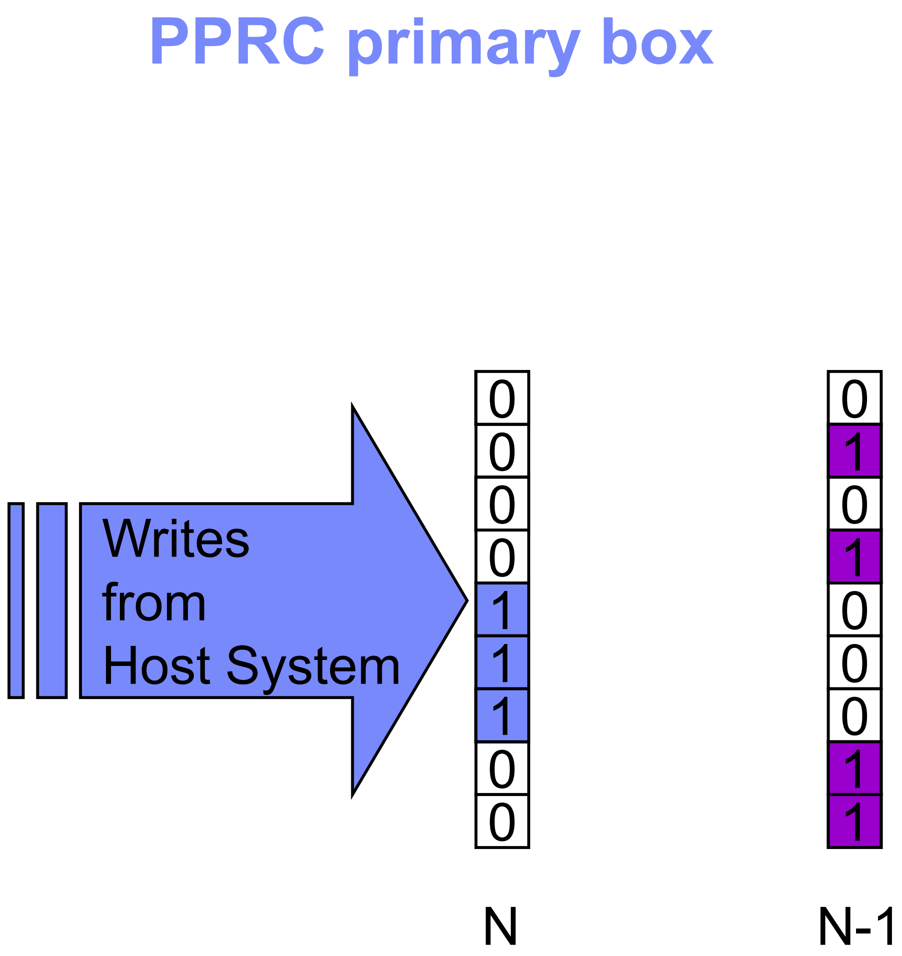
How does MGM Incremental Resynchronization work?



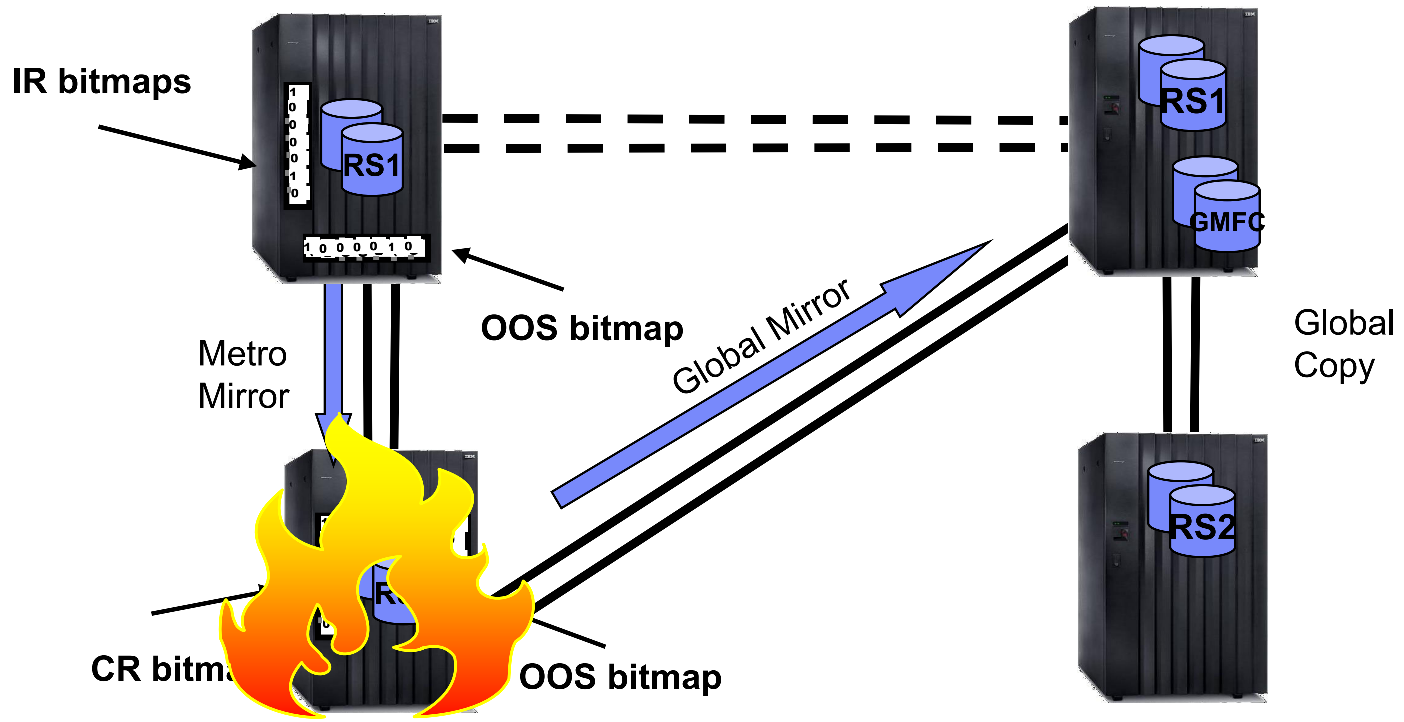
New writes still recorded in "N" bitmap

Issue queries to intermediate. If at least 3 CGs were successfully formed, and minimum time has passed:

Toggle N, N-1 bitmaps (N bitmap becomes zeros)

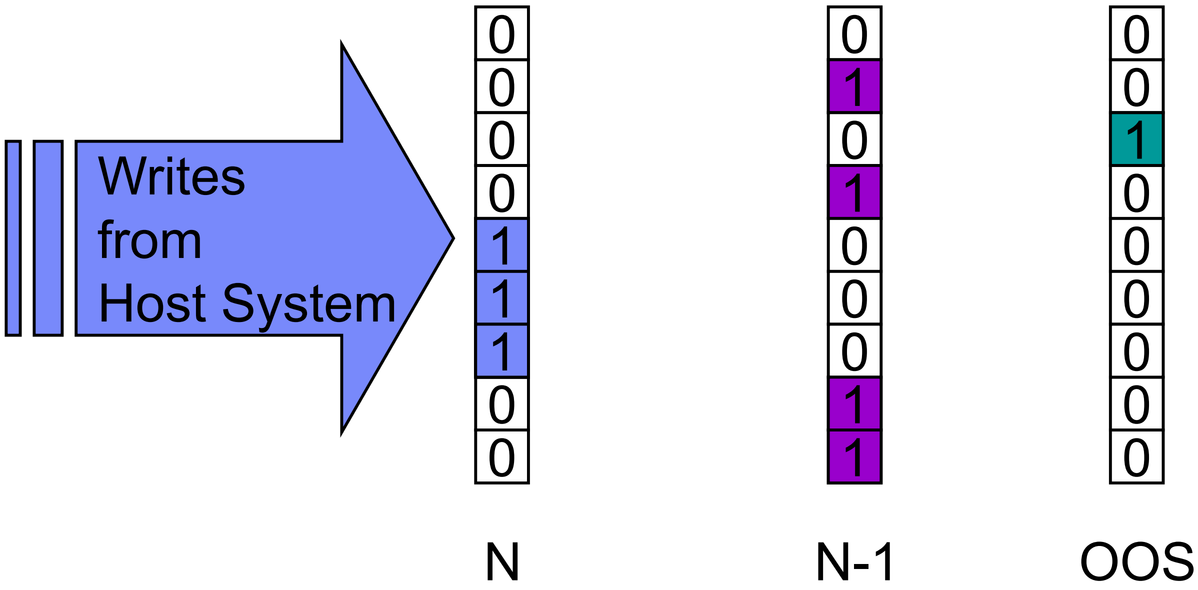
Continue recording writes in "N" bitmap

Incremental Resync – loss of intermediate site



How does MGM Incremental Resynchronization work?

PPRC primary box



OOS created when PPRC suspends

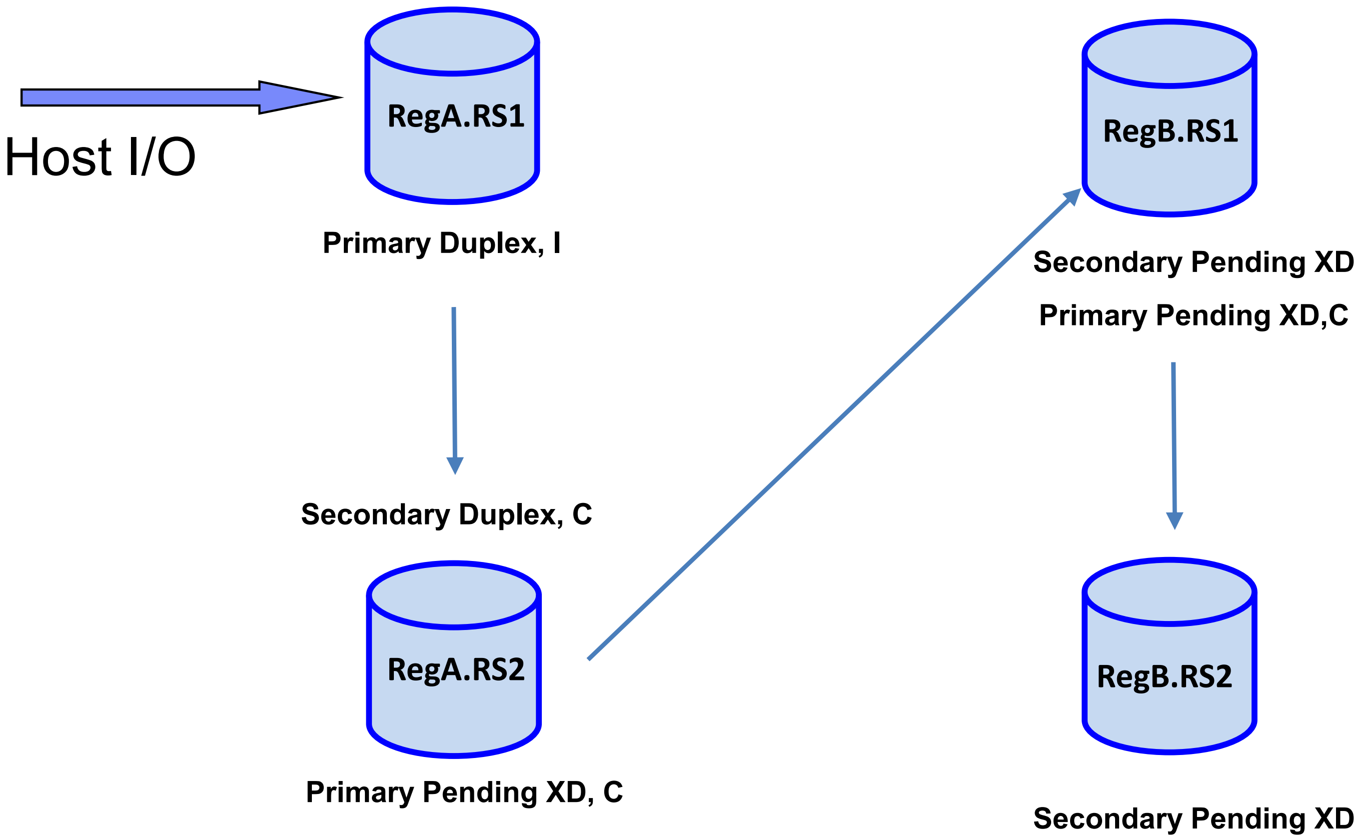
Merge N, N-1, and OOS bitmaps

Result is new OOS bitmap indicating what tracks need to be sent from the PPRC primary to tertiary site

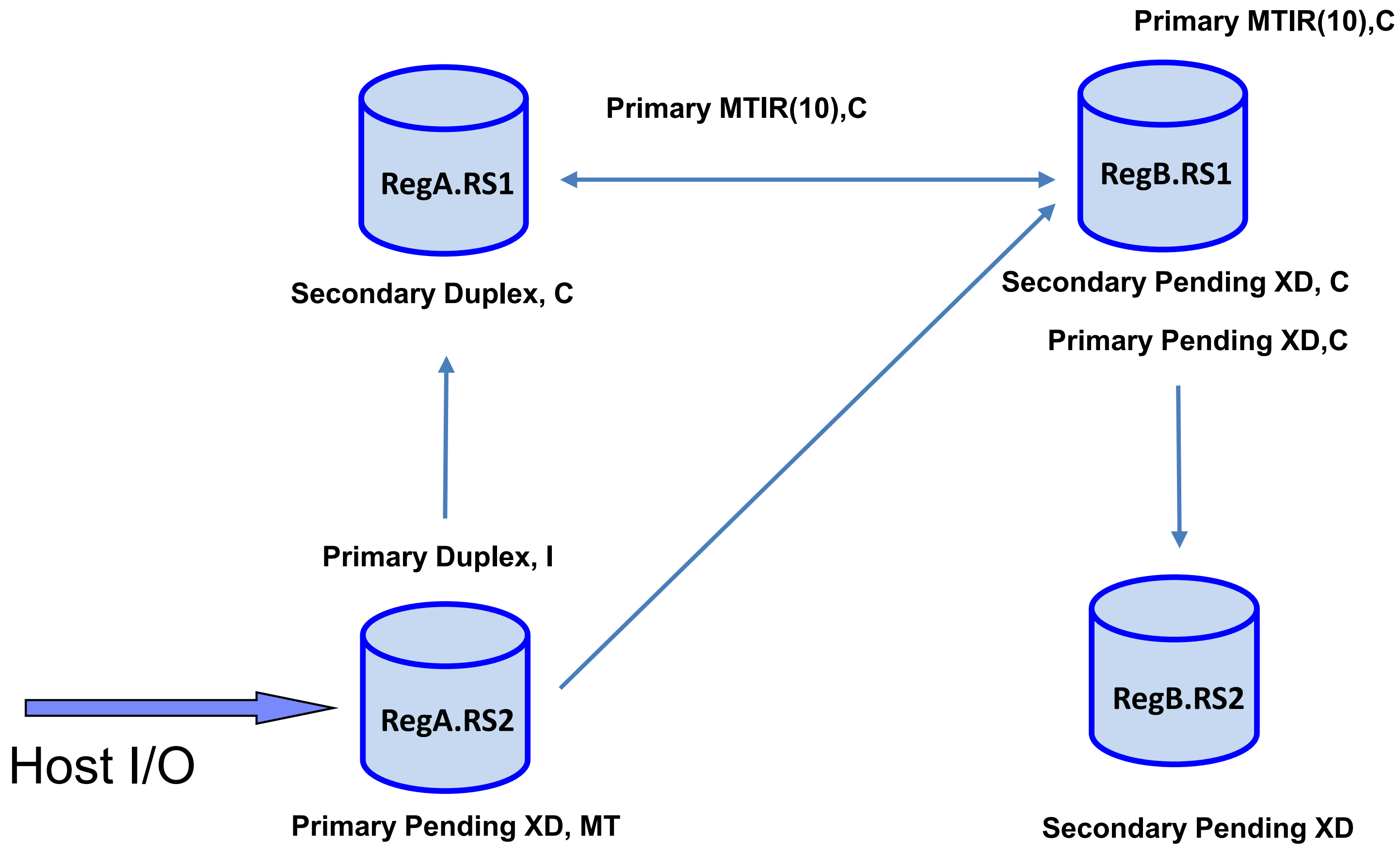
How does Multi-Target Incremental Resynchronization work?

- Multi-Target incremental resync is established by the DS8K microcode
- What is it?
  - A suspended relationship between two target devices that are being fed from the same source
  - Suspension code 10 is used to identify this relationship
  - Allows for the MTIR bitmap containing what tracks have been changed on these target devices to be used to incrementally resync that relationship even when one device is being fed synchronously, and the other asynchronously

Disk replication status – ‘cascade’ conditions



Disk replication status – ‘multi-target’ conditions



Disk replication status

- In an MGM environment the statuses returned by GDPS or CQUERY processing will be different from those in either a GDPS Metro or GDPS GM only configuration.
- In a ‘cascaded’ scenario
  - RegA.RS1 disk will be PRIMARY DUPLEX,I (Incremental Resync) in the GDPS Metro
  - RegA.RS1 disk will also (initially) be SIMPLEX to the RegB.RS1 disk (seen via toggle in Kg). Will appear as Primary MTIR(10),C after there has been a HyperSwap to the RegA.RS2 disk and a START SECONDARY to restore mirroring in Region A
  - RegA.RS2 disk will be SECONDARY DUPLEX,C (Cascade ON) in GDPS Metro
  - RegA.RS2 disk will be PRIMARY PENDING.XD,C in GDPS GM (Kg)
  - RegB.RS1 disk will be SECONDARY PENDING.XD in GDPS GM (Kr)
  - RegB.RS1disk will be PRIMARY PENDING.XD.C in GDPS Metro
  - RegB.RS1 will have no relationship to the RegA.RS1 in normal operation (Kr via toggle)
  - RegB.RS2 disk will be SECONDARY PENDING.XD in GDPS Metro
  - RegB.RS will have no relationship to any Reg.A disk in normal operation

Disk replication status

- In a ‘multi-target’ scenario (on RegA.RS2)
  - RegA.RS1 disk will be SECONDARY DUPLEX,C in GDPS Metro
  - RegA.RS1 disk will also be SIMPLEX to the RegB.RS1 disk (seen via toggle in Kg). Will appear as Primary MTIR(10),C after there has been a HyperSwap to the RegA.RS2 disk
  - RegA.RS2 disk will be PRIMARY DUPLEX,MT (Multi-target) in GDPS Metro
  - RegA.RS2 disk will be PRIMARY PENDING.XD,MT in GDPS GM (Kg)
  - RegB.RS1 disk will be SECONDARY PENDING.XD,C in GDPS GM (Kr)
  - RegB.RS1disk will be PRIMARY PENDING.XD.C in GDPS Metro
  - RegB.RS1 will have no relationship to the RegA.RS1 in normal operation (Kr via toggle)
  - RegB.RS2 disk will be SECONDARY PENDING.XD in GDPS Metro
  - RegB.RS will have no relationship to any Reg.A disk in normal operation
- Different statuses can exist depending on the scenario/situation. These will be explored in more detail during the lab exercises